Real-Time Fraud Detection System with Explainable AI & Live Dashboard

Summary Report

Problem Statement :

- Develop an AI-based Fraud Detection System to identify suspicious transactions in real time.
- Handle highly imbalanced financial data using advanced machine learning techniques.
- Use Explainable AI (SHAP) to make predictions transparent and trustworthy.
- Present insights through an interactive dashboard for fraud analysts and compliance teams.

Dataset :

Source : Kaggle

Name : IEEE-CIS Fraud Detection

-train_transaction.csv

-train_identity.csv

Technologies Used :

Programming & Data Processing

- Python, NumPy, Pandas

Machine Learning & Preprocessing

- Scikit-Learn, SMOTE, LightGBM, Optuna

Explainable AI

- SHAP (Explainable AI)

Deployment & Frontend

- Streamlit, Joblib

Data Visualization

- Matplotlib, Seaborn, Plotly

Workflow

1. **Data Cleaning & Preprocessing**
    - Merge transaction and identity datasets
    - Handle outliers using RobustScaler
2. **Class Imbalance Handling**
    - Apply SMOTE to balance fraud and non-fraud records
3. **Hyperparameter Optimization**
    - Use Optuna to find the best LightGBM parameters
4. **Model Training & Evaluation**
    - Evaluate using PR-AUC metric
    - Save trained model using Joblib

5. **Explainable AI Integration**
    - Use SHAP to explain fraud predictions
6. **Deployment & Dashboard**
    - Deploy real-time fraud detection system using Streamlit
    - Display predictions, risk scores, and visual insights interactively

Machine Learning Models & Techniques

Models Used

- LightGBM — Final selected model for high accuracy and fast performance
- XGBoost — Compared for performance evaluation
- Isolation Forest — Baseline anomaly detection model

Key Findings & Results :

Model Performance

- **LightGBM** achieved the best performance with **0.892 PR-AUC** and **84% fraud recall**
- Outperformed XGBoost and Isolation Forest in speed and accuracy
- Suitable for real-time fraud detection systems

Important Fraud Indicators (SHAP)

- **AmtToMeanRatio** — unusual transaction amounts
- **HourOfDay** — high risk during late-night hours
- **DeviceRisk** — suspicious or unknown devices

Risk Classification

- **Clear Tier** — safe transactions with 99.9% accuracy
- **Suspicious Tier** — moderate-risk transactions sent for MFA verification
- **Critical Tier** — top 2% highest-risk transactions containing most fraud cases

Business Impact

- Detected and prevented nearly **84% of fraudulent transactions**
- Reduced false alarms and operational costs
- Estimated annual savings: **$13.5 Million**
- Improved customer experience with faster legitimate checkouts

Class Imbalance Handling

- SMOTE used to generate synthetic fraud samples and balance the dataset

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Optimization & Evaluation

- Optuna used for automated hyperparameter tuning
- PR-AUC used as the main evaluation metric for fraud detection performance

Challenges Faced & Solutions :

1.Extreme Class Imbalance

- Fraud cases were only 3.5% of the dataset
- Solved using SMOTE, PR-AUC, and Recall-based evaluation

2. High-Cardinality Categorical Data

- Thousands of unique device and browser values caused memory issues
- Solved using frequency and risk-based encoding

3. Outliers in Transaction Amounts

- Large transaction variations affected model stability
- Solved using RobustScaler for better normalization

4. Black-Box Model Interpretability

- LightGBM predictions were difficult to explain
- Solved using SHAP for transparent and explainable AI predictions

Future Improvements :

1. Graph Neural Networks (GNNs):

- Detect fraud rings and connected suspicious accounts using graph-based analysis

2. Behavioral Biometrics:

- Integrate keystroke and mouse movement tracking to detect bots and fake users

3. Geospatial Fraud Detection:

- Implement location and travel-speed analysis to identify impossible transaction activities

Conclusion:

This project successfully developed an end-to-end AI-based Fraud Detection System using LightGBM, SHAP, and Streamlit. The model achieved a PR-AUC score of 0.892 with 84% fraud recall while effectively handling class imbalance and outliers using SMOTE and RobustScaler. The system provides accurate, explainable, and real-time fraud detection, with an estimated annual financial savings of $13.5 million.